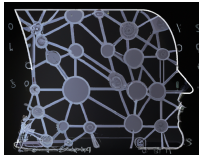
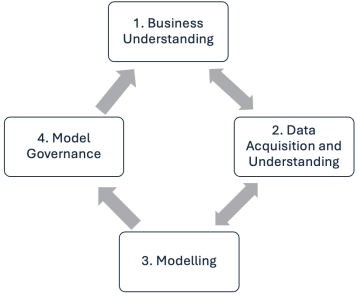
Applied Machine Learning



The Data Science Lifecycle
Main phases

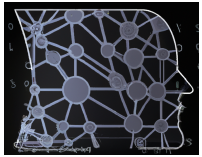


Learning goals

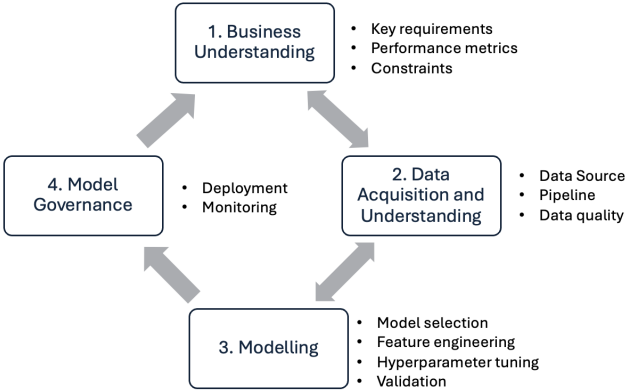
- Know phases of data science lifecycle
- Identify essential concepts, tools, and methods for each phase

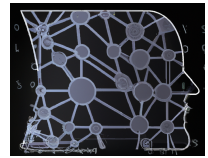
THE DATA SCIENCE LIFECYCLE

▸ "CRISP-DM" n.d.



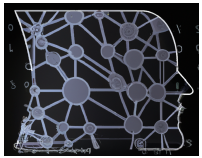
Lifecycle Management: Process of managing a (product) lifecycle from inception, through engineering, design and manufacturing to deployment and eventual disposal.
CRISP-DM (Cross-Industry Standard Process for Data Mining): A widely used framework that formalizes the iterative process of turning data into knowledge.

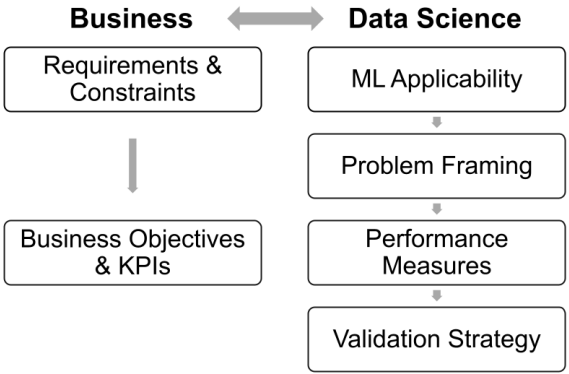




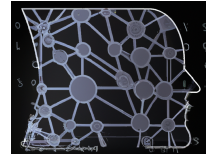
1. Business Understanding

OVERVIEW BUSINESS UNDERSTANDING





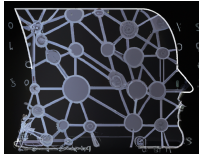
REQUIREMENTS & CONSTRAINTS



Key question: What is the end goal of the project?

- Define the intended **output**:
  - Report, dashboard, or interactive tool?
  - Scalable pipeline for production use?

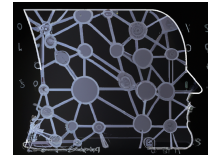
REQUIREMENTS & CONSTRAINTS



Key question: What is the end goal of the project?

- Define the intended **output**:
  - Report, dashboard, or interactive tool?
  - Scalable pipeline for production use?
- Identify constraints
  - Ethical: Avoid sensitive data use (e.g., address may proxy ethnicity)
  - Legal: Exclude protected attributes (e.g., gender in automated hiring)
  - Reproducibility: Ensure the analysis is fully reproducible in a single step
  - Explainability: All model decisions must be explainable to the user
  - Prediction Latency: Real-time predictions required
  - Model Size: Stay within specified memory constraints

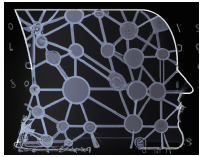
REQUIREMENTS & CONSTRAINTS



1. ML Applicability: Is ML justified?

- Explore non-ML baselines first (rules, heuristics, simple analytics)
- Use ML only if it adds clear value over simpler/existing alternatives

REQUIREMENTS & CONSTRAINTS



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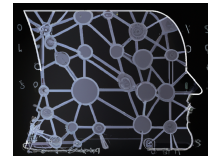
2. Problem Framing: How to frame the task?

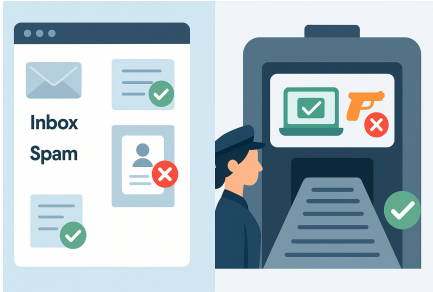
- *Imbalanced data:* classification vs. anomaly detection?
- *Pred. maintenance:* failure probability (classif.) vs. remaining lifetime (regr.)?
- Select the ML task that aligns with your KPI and data characteristics

Classification	Regression	Clustering
Anomaly Detection	Dimensionality Reduction	

BUSINESS OBJECTIVES: PERFORMANCE MEASURES



- **How to translate key performance indicators (KPIs) to metrics?**
  - **Spam detection**: Job application flagged as spam = unacceptable
    $\Rightarrow$ Occasional inbox spam is fine $\Rightarrow$ minimize FP $\Rightarrow$ maximize **precision**
  - **Airport security**: Weapon flagged as laptop = unacceptable
    $\Rightarrow$ Occasional false alarm is fine $\Rightarrow$ minimize FN $\Rightarrow$ maximize **recall**



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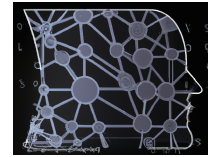

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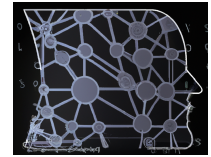
    $\Rightarrow$ Occasional false alarm is fine $\Rightarrow$ minimize FN $\Rightarrow$ maximize **recall**
- **Multiple objectives:** What if improving one metric worsens another?
  - *Scalarization*: Combine into one score (e.g., F_1)
  - *Pareto/frontier*: Analyze trade-off without collapsing (e.g., ROC curve)

True Values	**Predicted Values**		
	Pos	**Neg**	
Pos	TP	FN	**Sensitivity/Recall/TPR** $= \dfrac{TP}{TP + FN}$
Neg	FP	TN	**Specificity** $= \dfrac{TN}{TN + FP}$
	Precision $= \dfrac{TP}{TP + FP}$	**NPV** $= \dfrac{TN}{TN + FN}$	$\mathbf{F_1} = \dfrac{2\,\text{Precision Recall}}{\text{Precision} + \text{Recall}}$

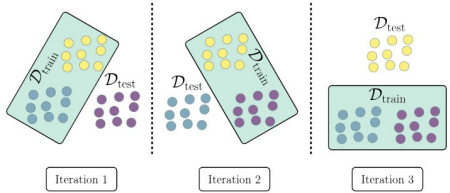
BUSINESS OBJECTIVES: VALIDATION STRATEGY



Key question: How to validate the model reliably and realistically?

1. Strategic Considerations

- **What should your model generalize to?**
  - *Blocking factors* (e.g., customers, hospitals, product types)
  - *Temporal effects* (e.g., seasonality, trends)
  - *Spatial effects* (e.g., regions, branches)
- **Simulate realistic deployment:**
  - **Time series:** Train on data from 2016–2020, deploy in 2021
  - **Validation:** Mimic deployment – train on 2016–2019, validate on 2020



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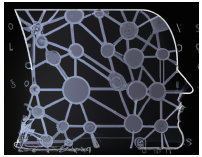

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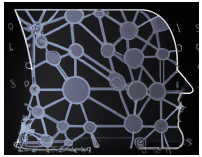
2. Beware of Data Leakage

- **Target Leakage:** Features leak future info not available at prediction time
  - Example: "Total Sales in Next Month" used to predict this month
- **Train-Test Leakage:** Information from the test set is used during training
  - Example: Scaling the full dataset before splitting

2. Data Acquisition and Understanding

DATA SOURCES AND DATA VERSIONING



Key question: What data is available and where does it come from?

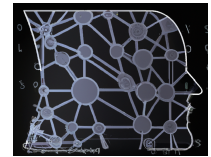
- **Internal vs. External:** External feeds may change format or be discontinued
- **Structured vs. Unstructured:** Tables vs. free-text, images, audio, etc.
- **Databases vs. Files:** DBs offer querying, consistency and security
- **On-Premise vs. Cloud:** Governed by regulations, cost and team expertise

Data Versioning:

- Data can evolve - use version control for data, like `git` for code
- Tools: `DVC` (Data Version Control), `Pachyderm`

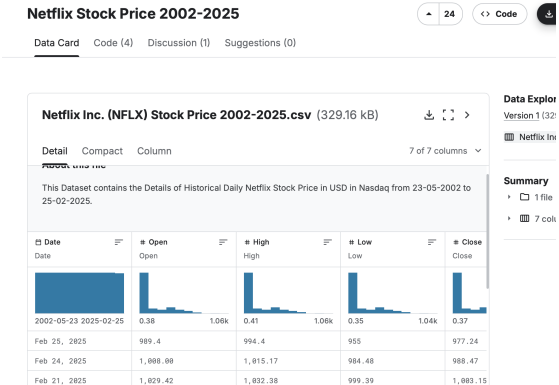
DATASHEETS

▸ "Gebru et al" 2018



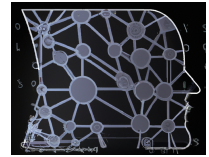
Datasheets summarize key metadata about a dataset's content, structure, provenance, and known limitations to support informed and responsible use.

- Why and how was the data collected?
- What are its limitations and ethical concerns?
- For which tasks is it suitable or unsuitable?



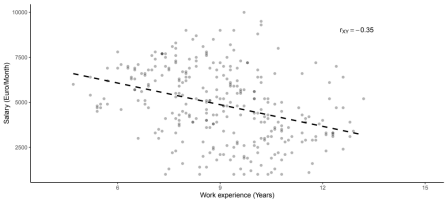
Source: ▸ "Kaggle Datasets" 2025

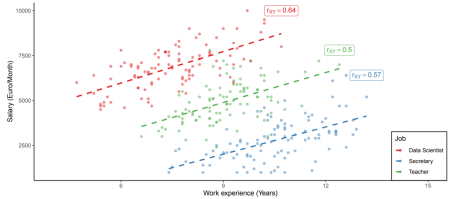
DATA EXPLORATION



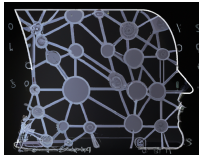
Key question: What features are in the data, and what do they reveal?

- **Build a data dictionary**:
  - Feature meaning, units, valid ranges/category-levels
  - Missing value encoding and special codes
- **Examine structure and quality to gain understanding**:
  - Compute (stratified) summary statistics and inspect missingness
  - Plot univariate and bivariate distributions
  - Identify outliers, anomalies, imbalances, redundancy, or high correlation
- **Watch for confounders:** unmeasured variables may distort associations





DATA QUALITY

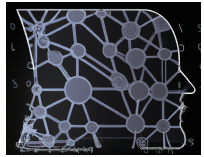


Key question: Is the data fit for modeling? *Garbage in, garbage out.*

- Assess missingness: How many values are missing, and where?
- Detect implausible or inconsistent values (requires domain expertise)
- Standardize formats (e.g., date/time parsing)
- Drop near-constant or non-informative features
- Remove redundancy and obsolete variables

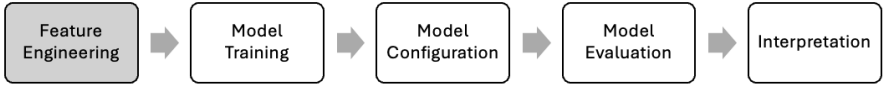
Watch for data bias:

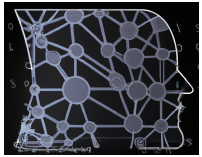
- *Sampling bias:* E.g., a survey at 10am in the city center misses working adults
- *Response bias:* Online reviews often reflect extreme opinions, not the median



3. Modeling

FEATURE ENGINEERING





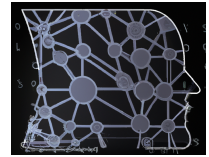
Feature Engineering transforms raw data into informative input variables for modeling.

- **Transformation**: Scaling, log-transform, one-hot encoding
- **Selection**: Wrapper and filter methods, model-based importance
- **Dimensionality Reduction**: PCA, t-SNE
- **Creation**: Aggregation, binning, interaction terms

Best Practices:

- Avoid overengineering: irrelevant features introduce noise
- Fit transformations **only on training data**; reuse parameters for test data
- Prevent leakage: never use test data in feature construction

MODEL TRAINING

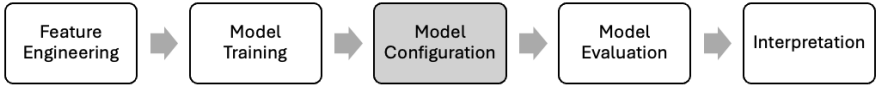


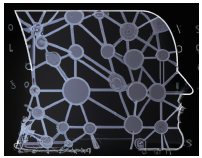
Feature Engineering → **Model Training** → Model Configuration → Model Evaluation → Interpretation

Treat model training like a scientific experiment:

- **Track all inputs precisely:**
  - Dataset version
  - Algorithm and hyperparameters
  - Code version (e.g., Git commit/tag)
- **Proceed iteratively:**
  - *Start simple* – build a fast, interpretable baseline
  - Gradually increase model complexity and validate each improvement

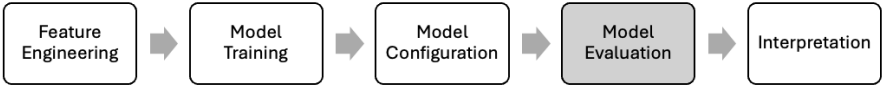
MODEL CONFIGURATION

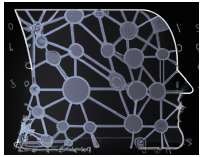




- How do you track the exact configuration of a model experiment?
- How do you record small changes between experiments?
- Store in plain text (e.g., `.yaml`, `.json`) for readability and version control
- Recommended tool: structured configuration managers
  - **Python**: "Hydra" n.d. - structured, hierarchical configuration management
  - **R**: "R config package" n.d. - environment-based YAML setup

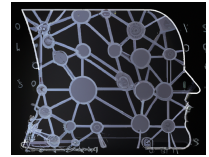
MODEL EVALUATION





- Select models based on predefined **metrics** (e.g., accuracy, AUC, RMSE)
- Use the **validation or test set** in line with your validation strategy
- Assess **robustness**: How does the model respond to noise, perturbations, or adversarial inputs?
  $\Rightarrow$ Add synthetic noise and measure changes in model parameters/performance

INTERPRETATION AND FAIRNESS



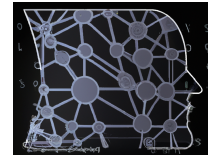
Feature Engineering → Model Training → Model Configuration → Model Evaluation → **Interpretation**

To understand and trust your model, apply methods from **Interpretable Machine Learning (IML)**:

- **Feature Importance**: Which features drive predictions globally?
- **Effects & Interactions**: Visualize marginal effects and feature interactions
- **Local Explanations**: Explain individual predictions
- **Bias Detection**: Reveal fairness issues and unintended discrimination
  - Example: Amazon's 2018 hiring tool penalized resumes with "women's" due to biased training data ▸ "The Guardian" 2018

MODEL CARDS FOR MODEL REPORTING

▸ "Mitchell et al." 2019



Model Card: Documentation of the model similar to a Datasheet for data

- **Model Details**: Algorithm, parameters, model version
- **Intended Use**: Intended use and out-of-scope use cases
- **Performance**: Key metrics, validation approaches
- **Analyses**: Feature/target distributions in train/test data
- **Ethical Considerations**: Failure modes, fairness concerns, inappropriate use

Model Card for Breast Cancer Wisconsin (Diagnostic) Dataset

Model Details

Overview

This model predicts whether breast cancer is benign or malignant based on image measurements.

Version

name: bba0bec9-9d5c-48fe-a291-72125c2c534a
date: 2020-09-25

Owners

- Model Cards Team, model-cards@google.com

References

- https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)
- https://minds.wisconsin.edu/bitstream/handle/1793/59692/TR1131.pdf

Considerations

Intended Users

- Medical professionals
- ML researchers

Use Cases

- Breast cancer diagnosis

Limitations

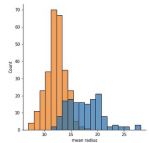
- Breast cancer diagnosis

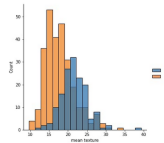
Ethical Considerations

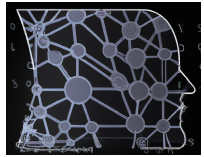
- Risk: Manual selection of image sections to digitize could create selection bias
  Mitigation Strategy: Automate the selection process

Train Set

426 rows with 30 features

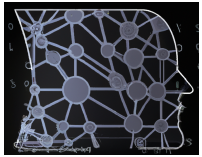






4. Model Governance

MODEL GOVERNANCE



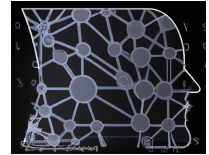
- Model Governance includes **people, processes, and technologies** needed to manage and protect your ML models.
- Ensures long-term maintainability and adaptation as requirements evolve.
- Two extremes must be avoided:
  - **Repression**: too many rules, rigid standards, excessive control
  - **Chaos**: too much speed, freedom, creativity, and unstructured change

Model governance steps are:

"Developing and deploying ML systems is relatively fast and cheap, but maintaining them over time is difficult and expensive."

▸ *"Hidden Technical Debt in ML Systems" n.d.*

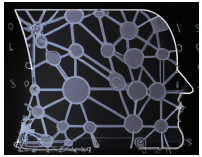
DEPLOYMENT



Deployment means making a model accessible to users (humans or other systems).

- Common deployment strategies:
  - **One-off**: Manually trained and deployed once; no regular updates
  - **Batch**: Periodically retrained on recent data (e.g., weekly updates)
  - **Real-time**: Continuously updated on streaming data; often latency-critical
- For frequent updates, establish model **versioning and lifecycle tracking**
  - Tools like ▸ "MLflow" n.d. provide version control and a model registry

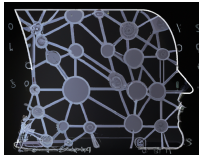
MONITORING



Monitoring ensures that deployed models work reliably and flags issues in production:

- **Pipeline Failures**: Bugs in data ingestion or transformation logic
- **Data Drift**: Changes in input distribution
  - Example: An image classifier trained on summer data fails when images contain winter snow
  - Also affects evolving targets (e.g., new categories or value ranges)
  - **Remedy**: Label sufficient new data and retrain the model
- **Concept Drift**: Changes in the meaning of labels
  - Example: Redefining “high blood pressure” from 140mmHg to 135mmHg
  - **Remedy**: Relabel historical data and retrain

USER FEEDBACK, REINFORCEMENT



Handover: At project completion, engage users and transfer responsibilities:

- **System Validation**: Confirm with stakeholders that the solution meet their needs
- **Project Handover**: Transfer system ownership to the production team

Feedback loops occur when model outputs influence future inputs:

- **Direct Loops**: A model affects the data it later learns from
  - Example: Netflix recommends certain movies, then only learns from user responses to those
- **Hidden Loops**: Multiple models indirectly affect each other via the environment
  - Example: Flash crashes triggered by interacting trading algorithms amplifying market reactions